

industries. These challenges will soon explode, since the Big Data created as a result of the billions of devices in use drastically increases security complexity. Both personal data (consumer driven) and Big Data (enterprise driven) are created by IoT devices. As consumers use more apps and IoT devices, more data is generated.

Today IoT devices need more storage capacity and IoT data needs to be stored in a cost-effective and secure manner.

It is important to understand the potential threats to an IoT device and add the appropriate defences while the system is being designed. Retrofitting security defences to a large network of IoT devices in the field is unfeasible, error prone and leaves customers at risk. Threat modelling can help development teams to understand what an attacker might do and why. Security must be provided to IoT architecture at the following zones:

- Device
- Field gateway
- Cloud gateway
- Services

What's trending

The Eclipse IoT community is today a pioneer in providing IoT security solutions. For example, Eclipse tinydtls provides an implementation of the DTLS (Datagram Transport Layer Security) protocol, ensuring transport layer security between the device and the server. Eclipse Keti provides an access control service that allows each stack in an IoT solution to protect its resources using a RESTful interface.

Today, most IoT systems are targets for cyber attacks. As a result, encryption is currently

gaining importance. Lightweight cryptography is an encryption method that features a small footprint and low computational complexity.

Future-proofing and a flexible approach are two important factors that IoT designers should consider. So they need to adopt 'security by design' principles. Future-proofing devices will soon become standard for IoT security. As malicious actors are constantly showing their skills at hacking networks, businesses must be flexible and proactive in their approach to security. 

References

- [1] <https://enterpriseiotinsights.com>>...
- [2] <https://www.itbusinessedge.com>
- [3] <https://techbeacon.com>> security
- [4] <https://iot.eclipse.org>
- [5] <https://www.electronicdesign.com>> secur..
- [6] Several other Web pages were referred to while writing this article.

By: Vinayak Ramachandra Adkoli

The author is a BE in industrial production. He has ten years of experience as a lecturer in the mechanical department of three different polytechnics. He is also a freelance writer and cartoonist.

OSFY Magazine Attractions During 2019-20

MONTH	THEME
March 2019	Mobile/Web App Development and Optimisation
April 2019	Cybersecurity, Open Source Firewall, Network Security and Monitoring
May 2019	AI, Deep Learning and Machine Learning
June 2019	DevOps Special
July 2019	Blockchain and Open Source
August 2019	Database Management and Optimisation
September 2019	Open Source and IoT
October 2019	Cloud Special: BigData, Hadoop, PaaS, SaaS, IaaS and Cloud
November 2019	Open Source on Windows
December 2019	Open Source Programming (Languages and Tools)
January 2020	Data Security, Storage and Backup
February 2020	Best in the World of Open Source (Tools and Services)